

Equation Sheet: Electrostatics

Intro Physics II • PHYS 235 (calc) • Knight ch. 22–25

fundamental = know cold (your starting points); **derived** = a line or two from the fundamentals; **bonus** = not required — might appear as an extra-credit problem only.

Symbols and constants

q, Q charge (C)	Φ electric flux (N·m ² /C)	U potential energy (J)
$\vec{E}$ electric field (N/C = V/m)	λ, η, ρ line / area / volume charge density	V potential (V = J/C)
r, d distance / separation (m)	A area	e 1.60×10^{-19} C
ϵ_0 8.85×10^{-12} C ² /(N·m ²),	$\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9$ N·m ² /C ²	
$\vec{r}_{12}$ separation vector, <i>points from charge 1 to charge 2</i> : $\vec{r}_{12} = \vec{r}_2 - \vec{r}_1$; $\hat{r}_{12} = \vec{r}_{12}/r_{12}$		

1. Coulomb's law & the electric field (Knight 22.4–22.5, 23.1–23.2)

fundamental

Coulomb's law, vector form
($\hat{r}_{12}$ points from charge 1 to charge 2; force *on* 2, *due to* 1; repulsive for like signs, $F \propto 1/r^2$)

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12}$$

Definition of the electric field

$$\vec{F} = q\vec{E}$$

Superposition of forces

$$\vec{F}_{\text{net}} = \sum_i \vec{F}_i$$

derived

Field of a point charge
($\hat{r}$ points from the charge to the field point)

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

Superposition of fields

$$\vec{E}_{\text{net}} = \sum_i \vec{E}_i$$

Coulomb's law, alternate vector form

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} q_1 q_2 \frac{\vec{r}_{12}}{r_{12}^3}$$

2. Continuous charge distributions (Knight 23.3–23.6)

fundamental

Chop into pieces, add up

$$\vec{E} = \sum_i \Delta \vec{E}_i, \quad \Delta E = \frac{1}{4\pi\epsilon_0} \frac{\Delta q}{r^2}$$

Charge densities

$$\lambda = \frac{Q}{L}, \quad \eta = \frac{Q}{A}, \quad \rho = \frac{Q}{V}$$

derived

Infinite line of charge (distance r)
(direction: radially outward from the line, for $\lambda > 0$)

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

Infinite plane / sheet of charge
(direction: perpendicular to the sheet, away from it on both sides, for $\eta > 0$)

$$E = \frac{\eta}{2\epsilon_0} \quad (\text{uniform — no } r)$$

Ring of charge Q , radius R , on axis at z
(direction: along the axis, away from the ring for $Q > 0$)

$$E = \frac{1}{4\pi\epsilon_0} \frac{Qz}{(z^2 + R^2)^{3/2}}$$

Disk of charge Q , radius R , on axis at z
($\eta = Q/\pi R^2$; direction: along the axis, away from the disk for $Q > 0$; $R \rightarrow \infty$ recovers the infinite sheet; $z \gg R$ recovers a point charge)

$$E = \frac{\eta}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right)$$

3. Conductors in equilibrium (Knight 24.4–24.6)

fundamental

Field inside the conducting material	$\vec{E} = 0$
Excess charge	all on the outer surface
Field at the outer surface	$\perp$ to the surface, $E = \frac{\eta}{\epsilon_0}$

derived

Empty cavity inside a conductor	$\vec{E} = 0$ in the cavity (Faraday shielding)
Charge $+q$ placed in the cavity	$-q$ on the inner wall, $+q$ on the outer surface

4. Electric potential energy & potential (Knight 25.1–25.7)

fundamental

Two point charges (keep the signs; $U \rightarrow 0$ as $r_{12} \rightarrow \infty$; one power of r)	$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$
Energy conservation	$K_i + U_i = K_f + U_f, \quad W_{\text{elec}} = -\Delta U$
Definition of potential	$U = qV \quad (1 \text{ V} = 1 \text{ J/C})$
Point charge (scalar; keep the sign; $V \rightarrow 0$ as $r \rightarrow \infty$)	$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}, \quad V_{\text{net}} = \sum_i \frac{1}{4\pi\epsilon_0} \frac{q_i}{r_i}$

derived

Three or more charges: sum over every <i>pair</i> (the work to assemble the whole arrangement — order doesn't matter)	$U_{\text{total}} = \frac{1}{4\pi\epsilon_0} \sum_{\text{pairs } (i,j)} \frac{q_i q_j}{r_{ij}}$
Equipotentials	$\perp$ field lines; $\vec{E}$ points toward decreasing V
Potential in a uniform field (d measured along $\vec{E}$, from initial to final point)	$V_f - V_i = -Ed$

5. Electric dipoles (Knight 23.7, 25.3)

fundamental

Dipole moment ($\vec{d}$ points from $-q$ to $+q$)	$\vec{p} = q\vec{d}$
Torque on a dipole in a uniform field	$\vec{\tau} = \vec{p} \times \vec{E}$
Potential energy of a dipole in a uniform field (most stable, i.e. minimum U , when $\vec{p} \parallel \vec{E}$)	$U = -\vec{p} \cdot \vec{E}$

6. Electric flux & Gauss's law (Knight 24.1–24.3)

fundamental

Electric flux through a surface

$$\Phi = \oint \vec{E} \cdot d\vec{A}$$

Gauss's law

(any closed surface; only the *enclosed* charge matters)

$$\Phi = \frac{Q_{\text{enc}}}{\epsilon_0}$$

derived

Uniform field through a flat area

$$\Phi = EA \cos \theta \quad (\theta \text{ from the normal})$$

Charge *outside* a closed surface

$$\Phi_{\text{net}} = 0$$

Sphere / spherical shell, total charge Q
($r > R$)
($\hat{r}$ points from the center outward)

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r} \quad (\text{same as a point charge})$$

7. Doing the calculus: continuous distributions & Gauss's law (Knight 23.3–23.6, 24.1–24.3, 25.4)

fundamental

Potential of a continuous distribution — chop, add up
(V is a scalar: no components, no cancellation to track)

$$V = \int dV, \quad dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r}$$

Evaluating a Gaussian-surface integral
(pick a surface where E is constant and either $\parallel$ or $\perp$ to $d\vec{A}$ everywhere)

$$\oint \vec{E} \cdot d\vec{A} = E \oint dA = EA \Rightarrow E = \frac{Q_{\text{enc}}}{\epsilon_0 A}$$

derived — electric field

Finite line of charge Q , length L , at perpendicular distance y from its midpoint

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{y z_0}$$

($z_0 = \sqrt{y^2 + (L/2)^2}$; direction $\perp$ to the line;
 $L \rightarrow \infty$ recovers the infinite-line result above)

derived — electric potential

Ring of charge Q , radius R , on axis at z

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{z^2 + R^2}}$$

Disk of charge Q , radius R , on axis at z
($\eta = Q/\pi R^2$)

$$V = \frac{\eta}{2\epsilon_0} \left(\sqrt{z^2 + R^2} - z \right)$$

8. Bonus: how $\vec{F}$, U , $\vec{E}$, V connect

might appear as an extra-credit problem only

Force $\leftrightarrow$ energy

Force is the (negative) slope of U

$$F_x = -\frac{dU}{dx}, \quad \vec{F} = -\vec{\nabla}U$$

U is (minus) the work integral of $\vec{F}$
(same statement as $W_{\text{elec}} = -\Delta U$ above, written
as a line integral)

$$U_f - U_i = -\int_i^f \vec{F} \cdot d\vec{s}$$

Field $\leftrightarrow$ potential

Field is the (negative) slope of V

$$E_x = -\frac{dV}{dx}, \quad \vec{E} = -\vec{\nabla}V$$

V is (minus) the line integral of $\vec{E}$

$$V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s}$$

The two rows are the same relationship twice: U and V are what you integrate a force/field *against* to get; $\vec{F}$ and $\vec{E}$ are what falls out when you differentiate U or V . Swap $U \rightarrow V$ and $\vec{F} \rightarrow \vec{E}$ (divide through by q) and the left column becomes the right column.

$\vec{\nabla} \equiv \frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}$ — the gradient operator, in Cartesian coordinates.

Companion simulations at hoppese.github.io/Intro-physics-sims/: `electroscope`,
`e-fields-conductors-and-insulators`, `electrostatics-explorer`, `field-superposition-explorer`,
`discrete-charge-builder-3d`, `gauss-flux`, `gauss-applications`, `assembling-charges`, `pe-vs-separation`,
`equipotential-explorer`.